

	Data Sheet Grid Code Variant 69	2002181EN Rev. 0 / 19.07.2019
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2.2 Protection Relay

The grid operator may require that every individual WT is equipped with a protection relay to ensure safety for the grid. Nordex offers an option for the implementation of a generally suitable relay. The following settings are chosen partly according to the grid code requirements (over frequency protection stages 1 and 2) and mainly as default settings. Some protection requirements by the grid operator may limit the possible operating ranges of the WT.

Table 4: Protection relay settings

Function	Response threshold	Delay time
Over voltage protection stage 1	$U > 1.15 U_{nom}$	60 s
Over voltage protection stage 2	$U > 1.25 U_{nom}$	0.1 s
Under voltage protection stage 1	$U < 0.87 U_{nom}$	180 s
Under voltage protection stage 2	$U < 0.80 U_{nom}$	11 s
Over frequency protection stage 1	$f > 52.0 \text{ Hz}$	0.2 s
Over frequency protection stage 2	$f > 52.0 \text{ Hz}$	0.2 s
Under frequency protection stage 1	$f < 47.5 \text{ Hz}$	300 s
Under frequency protection stage 2	$f < 47.0 \text{ Hz}$	0.2 s
Frequency gradient protection	4 Hz/s	1 s

2.3 Fault-Ride-Through

The WT can ride through transient voltage dips and swells without disconnection as shown in the voltage-time-diagram (Figure 1).

The dashed line indicates the grid code's fault-ride-through (FRT) requirement at the wind farm's point of connection. The WT will remain connected as long as all voltages at the low voltage terminal stay within the solid lines. The same minimum short-term voltage-time-capabilities can be realized for the medium voltage terminals and the grid connection point, since the WT provides voltage support. The exact capabilities depend on the wind farm layout, the OLTC position and the chosen grid support settings.